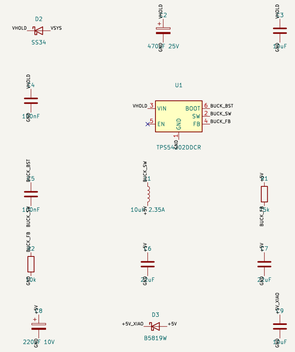
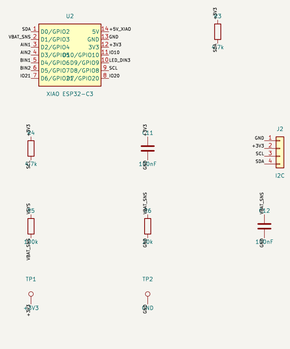
1 Battery input (2S / 3S LiPo), 20 A blade fuse, power switch, surge clamp + ESC power



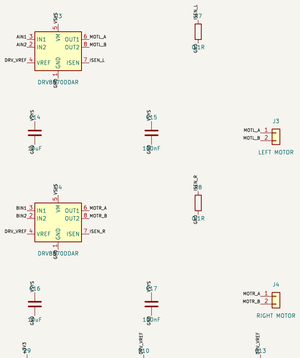
2 Logic power (TPS54202), XIAO feed



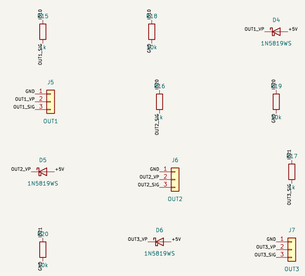
3 XIAO ESP32-C3, I2C, battery sense



4 2x DRV8870 drive, 2.5 A current limit



5 PWM outputs (ESC or servo)



6 RGB status LED (WS2812B strip piece)



Power flags (ERC)



Cardboard Bot v2 main board — rev B — Columbia Gadget Works
Generated from hardware/gen/circuit.py (do not hand-edit; edit circuit.py and regenerate).
All connections are made by net labels at pin ends.
Power: 2S or 3S LiPo on J1. Everything on VSYS is rated for 3S (12.6 V) with margin; SMAJ13A clamps spikes at about 21 V.
XIAO pins: D0 SDA(GPIO2) D1 VBAT/6(GPIO3) D2–D3 left DRV8870 IN1/IN2 D4–D5 right DRV8870 IN1/IN2 D6 OUT3(GPIO21) D7 OUT2(GPIO20) D8 SCL(GPIO8) D9 LED(GPIO9) D10 OUT1(GPIO10)
10k pull-downs on every motor input and output keep motors and ESCs off while the ESP32 boots.

Columbia Gadget Works	
Sheet: /	
File: cardboard-bot.kicad_sch	
Title: Cardboard Bot v2 main board	
Size: A1	Date: 2026-10-30
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Rev: B
	Id: 1/1